



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 7

Team No.: 25

Project Title: Linux Service and Process Supervision

Team Members

Roll Number	Student Name	Signature
2520030589	B.Niharika	
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2520030302	V.Satish	

1. Abstract

(Summarize the project idea, problem background, relevant Operating Systems/System Programming concepts, proposed Linux-based solution, and expected outcome. The abstract should not exceed 250 words.)

The *Linux Service and Process Supervision* project is designed to monitor, manage, and supervise processes and services running on a Linux operating system. The system provides a simple way to observe active processes, monitor CPU and memory usage, and manage system services efficiently.

The project enables users to view running processes along with important details such as Process ID (PID), process state, CPU usage, and memory consumption. It also provides functionalities to start, stop, and restart Linux services and monitor their current status. The system can identify resource-intensive processes and detect services that are inactive or have stopped unexpectedly.

This project demonstrates important *Operating System concepts*, including process management, process states, CPU utilization, memory management, service management, and system monitoring. Linux system commands and utilities are used to collect process information and perform service-management operations.

The proposed system helps improve system visibility and simplifies basic process and service administration. It also provides practical knowledge of how Linux handles background services and running processes, making it useful for learning and understanding real-world operating system management.

2. Problem Statement

(Clearly describe the problem that the project intends to address and its relevance to Operating Systems and Systems Programming.)

Managing and monitoring multiple processes and system services in a Linux environment can be difficult when relying only on individual command-line utilities. Users may face challenges in identifying resource-intensive processes, detecting inactive or failed services, and controlling services efficiently. The project aims to develop a *Linux Service and Process Supervision* system that provides a centralized and simple method to monitor process activity, CPU and memory usage, process states, and service status, while supporting basic service control operations. This project is relevant to **Operating Systems and Systems Programming** because it applies concepts such as process management, process states, CPU and memory utilization, service management, system calls, and Linux system utilities to solve a practical system-administration problem.

3. Objectives

(List the specific objectives of the project.)

1. To monitor and display the status and details of running processes in a Linux operating system.
2. To monitor CPU and memory utilization and identify processes consuming excessive system resources.
3. To manage Linux services by providing operations such as starting, stopping, restarting, and checking service status.
4. To apply Operating System and Systems Programming concepts such as process management, system calls, process states, and Linux system utilities.

4. Proposed Methodology

(Briefly explain how the project will be designed and implemented in Linux. Describe the major functionality, approach, and flow of the proposed solution.)

*The proposed system will be designed and implemented in a Linux environment using system commands, shell scripting, and Linux system utilities to monitor and manage processes and services. The system will collect process information such as PID, process state, CPU usage, and memory usage from Linux system interfaces such as `/proc`. Service information will be obtained using Linux service-management utilities such as `systemctl`. The system will analyze the collected information and display the current process and service status to the user. Users can select a process or service and perform operations such as monitoring, starting, stopping, or restarting services. The overall flow involves **collecting system information → processing and analyzing data → displaying status → performing user-requested management operations → continuously monitoring the system**. This approach demonstrates practical concepts of process management, service supervision, resource monitoring, system calls, and Linux systems programming.*

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project.
Process Management	Monitors and manage running processes.
Process States	Identify running, sleeping, stopped, and terminated processes.
fork()/exec()	Create and execute processes.
wait()/waitpid()	Monitor child processes and their termination.
kill()/Signals	Control or terminate processes using signals.
/proc&systemctl	Collect process information and manage Linux services.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and testing the project.
C / C++	Used to implement process and service supervision functionality.
GCC Compiler	Used to compile and execute C/C++ programs in Linux.
Linux SystemCalls/POSIX APIs	Used for process creation, control, synchronization, and monitoring.
systemd/systemctl	Used to start,stop,restart,and monitor Linux services.

8. Expected Outcome

(Describe the expected functionality, output, and outcome of the project.)

The project is expected to provide a simple Linux-based system for monitoring and supervising running processes and system services.

It will display process details such as PID, process state, CPU usage, and memory usage, along with the current status of Linux services.

The system will allow users to start, stop, restart, and monitor services and identify processes consuming excessive system resources.

The project will demonstrate practical implementation of process management, service management, system calls, and Linux operating system concepts.

GitHub Repository Link: _____

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____